PROJECT TITLE:

Digital Payment & Financial Inclusion in Africa

CASE STUDY

Scope :

- Examines the adoption of digital payments across selected countries
- Focuses on the relationship between *internet penetration, GDP, and digital financial inclusion*.

Data Source :

- World Bank Open Data (Digital Payments, Internet Users, GDP).

Ojectives :

- Identify trends in digital payment adoption.
- Understand how digital infrastructure influences economic growth.
- Highlight emerging economies showing rapid adotion.

EXECUTIVE SUMMARY

This project analyzes the adoption of digital payments across multiple countries using datasets from the World Bank. By examining key indicators; digital payments (% of population age 15+), internet penetration, and GDP. This analysis identifies patterns and relationships that highlight the role of digital infrastructure in economic development.

The case study focuses on comparing emerging and developed economies, revealing that countries with higher internet penetration rates tend to adopt digital payments more rapidly. Additionally, the analysis shows that some emerging economies are accelerating in digital finance adoption, narrowing the gap with developed countries.

Through data cleaning, transformation, and visualization, the project presents clear insights into how financial technology adoption is intertwined with economic growth and inclusion. The findings are illustrated with professional line charts and comparative visualizations, providing a compelling narrative of the global digital payment landscape.

This project serves as both a data analytics portfolio piece and a foundation for scholarship applications, demonstrating the ability to combine technical skills with real-world economic insight.

AUTHOR

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In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

In [2]:
```
digital_file = r"C:\Users\JustDamad\Desktop\data_folder\WB_GS_G20_T_WIDEF.csv"
internet_file = r"C:\Users\JustDamad\Desktop\data_folder\WB_WDI_IT_NET_USER_ZS_WIDE
gdp_file = r"C:\Users\JustDamad\Desktop\data_folder\WB_WDI_NY_GDP_MKTP_CD_WIDEF.csv
```

In [3]:
```
## Let's load the data

digital = pd.read_csv(digital_file)
internet = pd.read_csv(internet_file)
gdp = pd.read_csv(gdp_file)
```

In [4]:
```
## Let's preview the data

print(digital.head())
print(internet.head())
print(gdp.head())
```

```
  FREQ FREQ_LABEL REF_AREA REF_AREA_LABEL    INDICATOR  \
0    A     Annual      AFG    Afghanistan  WB_GS_G20_T
1    A     Annual      AFG    Afghanistan  WB_GS_G20_T
2    A     Annual      AFG    Afghanistan  WB_GS_G20_T
3    A     Annual      AGO         Angola  WB_GS_G20_T
4    A     Annual      AGO         Angola  WB_GS_G20_T

                                   INDICATOR_LABEL SEX SEX_LABEL  \
0  Made or received digital payments in the past ...   F    Female
1  Made or received digital payments in the past ...   M      Male
2  Made or received digital payments in the past ...  _T     Total
3  Made or received digital payments in the past ...   F    Female
4  Made or received digital payments in the past ...   M      Male

  UNIT_MEASURE UNIT_MEASURE_LABEL  ... UNIT_MULT UNIT_MULT_LABEL  OBS_STATUS  \
0           PT         Percentage  ...         0           Units           A
1           PT         Percentage  ...         0           Units           A
2           PT         Percentage  ...         0           Units           A
3           PT         Percentage  ...         0           Units           A
4           PT         Percentage  ...         0           Units           A

  OBS_STATUS_LABEL OBS_CONF OBS_CONF_LABEL   2014   2017   2021  2022
0     Normal value       PU         Public   1.12   4.08   3.96   NaN
1     Normal value       PU         Public   9.79  17.49  11.94   NaN
2     Normal value       PU         Public   5.57  10.83   7.88   NaN
3     Normal value       PU         Public  18.39    NaN    NaN   NaN
4     Normal value       PU         Public  31.53    NaN    NaN   NaN

[5 rows x 22 columns]
       STRUCTURE                STRUCTURE_ID ACTION FREQ REF_AREA  \
0  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      KGZ
1  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      NCL
2  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      LAC
3  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      GBR
4  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      CHL

               INDICATOR SEX AGE URBANISATION UNIT_MEASURE  ...     2015  \
0  WB_WDI_IT_NET_USER_ZS  _T  _T           _T       PT_POP  ...  30.2470
1  WB_WDI_IT_NET_USER_ZS  _T  _T           _T       PT_POP  ...  74.0018
2  WB_WDI_IT_NET_USER_ZS  _T  _T           _T       PT_POP  ...  53.3000
3  WB_WDI_IT_NET_USER_ZS  _T  _T           _T       PT_POP  ...  92.0003
4  WB_WDI_IT_NET_USER_ZS  _T  _T           _T       PT_POP  ...  76.6296

      2016     2017    2018     2019     2020     2021     2022     2023  2024
0  40.5000  50.8000  62.800  64.1000  72.0000  77.1177  82.5991  88.4701   NaN
1      NaN  82.0058     NaN      NaN      NaN      NaN      NaN      NaN   NaN
2  56.4000  59.2000  62.600  67.9000  73.3000  75.3000  76.6000  79.7000  81.4
3  94.7758  90.4246  90.692  92.5166  94.8182  96.1965  96.2989  96.2988   NaN
4  83.5586  82.3275  84.900  85.0176  87.4593  90.2328  93.5377  94.4574   NaN

[5 rows x 76 columns]
       STRUCTURE                STRUCTURE_ID ACTION FREQ REF_AREA  \
0  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      MRT
1  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      SYR
2  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      LMY
3  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      CYP
```

```
4  datastructure  WB.DATA360:DS_DATA360(1.3)      I    A      MAR

               INDICATOR SEX AGE URBANISATION UNIT_MEASURE  ...          2015  \
0  WB_WDI_NY_GDP_MKTP_CD  _T  _T           _T          USD  ...  6.166864e+09
1  WB_WDI_NY_GDP_MKTP_CD  _T  _T           _T          USD  ...  1.646686e+10
2  WB_WDI_NY_GDP_MKTP_CD  _T  _T           _T          USD  ...  2.545729e+13
3  WB_WDI_NY_GDP_MKTP_CD  _T  _T           _T          USD  ...  1.990928e+10
4  WB_WDI_NY_GDP_MKTP_CD  _T  _T           _T          USD  ...  1.104138e+11

           2016          2017          2018          2019          2020  \
0  6.398748e+09  6.800136e+09  7.473551e+09  7.894765e+09  8.260752e+09
1  1.259785e+10  1.636984e+10  2.149778e+10  2.258305e+10  1.204775e+10
2  2.572274e+13  2.812368e+13  2.994595e+13  3.086348e+13  2.988266e+13
3  2.104646e+10  2.294657e+10  2.575400e+10  2.619667e+10  2.555509e+10
4  1.115729e+11  1.185406e+11  1.273411e+11  1.289203e+11  1.213536e+11

           2021          2022          2023          2024
0  9.222536e+09  9.557967e+09  1.072684e+10  1.090805e+10
1  1.435321e+10  2.362283e+10           NaN           NaN
2  3.523542e+13  3.696530e+13  3.805933e+13  3.939710e+13
3  3.037264e+10  3.121804e+10  3.507542e+10  3.763453e+10
4  1.420221e+11  1.312453e+11  1.460361e+11  1.606110e+11

[5 rows x 106 columns]
```

In [5]:
```
## Since we want to do analysis on African Countries, let's filter fot African Coun

africa_countries = [
    "Nigeria", "Kenya", "Ghana", "South Africa", "Egypt",
    "Morocco", "Uganda", "Tanzania", "Rwanda"
]

digital_africa = digital[digital["REF_AREA_LABEL"].isin(africa_countries)]
internet_africa = internet[internet["REF_AREA_LABEL"].isin(africa_countries)]
gdp_africa = gdp[gdp["REF_AREA_LABEL"].isin(africa_countries)]
```

In [6]:
```
# Let's List all columns that look like years

year_columns = [col for col in digital_africa.columns if col.isdigit()]
print(year_columns)
```

```
['2014', '2017', '2021', '2022']
```

In [7]:
```
digital_africa = digital_africa[["REF_AREA_LABEL", "2022"]].rename(columns = {"2022
internet_africa = internet_africa[["REF_AREA_LABEL", "2022"]].rename(columns = {"20
gdp_africa = gdp_africa[["REF_AREA_LABEL", "2022"]].rename(columns = {"2022" : "GDP
```

In [8]:
```
## Let's Strip spaces from column names

digital.columns = digital.columns.str.strip()
internet.columns = internet.columns.str.strip()
gdp.columns = gdp.columns.str.strip()

## Let's Filter for African countries

africa_countries = [
```

```
    "Nigeria", "Kenya", "Ghana", "South Africa", "Egypt",
    "Morocco", "Uganda", "Tanzania", "Rwanda"
]

digital_africa = digital[digital['REF_AREA_LABEL'].isin(africa_countries)]
internet_africa = internet[internet['REF_AREA_LABEL'].isin(africa_countries)]
gdp_africa = gdp[gdp['REF_AREA_LABEL'].isin(africa_countries)]
```

In [9]:
```
## Let's Identify most recent year column available

digital_years = [col for col in digital_africa.columns if col.isdigit()]
internet_years = [col for col in internet_africa.columns if col.isdigit()]
gdp_years = [col for col in gdp_africa.columns if col.isdigit()]

print("Digital Payments Years:", digital_years)
print("Internet Users Years:", internet_years)
print("GDP per Capita Years:", gdp_years)
```

```
Digital Payments Years: ['2014', '2017', '2021', '2022']
Internet Users Years: ['1990', '1991', '1992', '1993', '1994', '1995', '1996', '199
7', '1998', '1999', '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
'2008', '2009', '2010', '2011', '2012', '2013', '2014', '2015', '2016', '2017', '201
8', '2019', '2020', '2021', '2022', '2023', '2024']
GDP per Capita Years: ['1960', '1961', '1962', '1963', '1964', '1965', '1966', '196
7', '1968', '1969', '1970', '1971', '1972', '1973', '1974', '1975', '1976', '1977',
'1978', '1979', '1980', '1981', '1982', '1983', '1984', '1985', '1986', '1987', '198
8', '1989', '1990', '1991', '1992', '1993', '1994', '1995', '1996', '1997', '1998',
'1999', '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007', '2008', '200
9', '2010', '2011', '2012', '2013', '2014', '2015', '2016', '2017', '2018', '2019',
'2020', '2021', '2022', '2023', '2024']
```

In [10]:
```
## Let's Use the most recent year available

digital_latest_year = digital_years[-1]
internet_latest_year = internet_years[-1]
gdp_latest_year = gdp_years[-1]
```

In [11]:
```
## to get a better data, let's reduce it to Country and Latest Year

digital_latest = digital_africa[["REF_AREA_LABEL", digital_latest_year]].rename(col
internet_latest = internet_africa[["REF_AREA_LABEL", internet_latest_year]].rename(
gdp_latest = gdp_africa[["REF_AREA_LABEL", gdp_latest_year]].rename(columns = {gdp_
```

In [12]:
```
## Because of the empty digital column, we might have duplicates
## Let's drop duplicates

digital_latest = digital_latest.drop_duplicates(subset = ["REF_AREA_LABEL"])
internet_latest = internet_latest.drop_duplicates(subset = ["REF_AREA_LABEL"])
gdp_latest = gdp_latest.drop_duplicates(subset = ["REF_AREA_LABEL"])
```

In [13]:
```
## Let's merge the DATASETS

df = digital_africa.merge(internet_africa, on = "REF_AREA_LABEL")
df = df.merge(gdp_africa, on = "REF_AREA_LABEL")
```

```
## Let's rename the columns for clarity

df = df.rename(columns = {"REF_AREA_LABEL" : "Country"})

df.head()
```

Out[13]:

	FREQ_x	FREQ_LABEL_x	REF_AREA_x	Country	INDICATOR_x	INDICATOR_LABEL_x	SEX_
0	A	Annual	GHA	Ghana	WB_GS_G20_T	Made or received digital payments in the past ...	
1	A	Annual	GHA	Ghana	WB_GS_G20_T	Made or received digital payments in the past ...	N
2	A	Annual	GHA	Ghana	WB_GS_G20_T	Made or received digital payments in the past ...	_
3	A	Annual	KEN	Kenya	WB_GS_G20_T	Made or received digital payments in the past ...	
4	A	Annual	KEN	Kenya	WB_GS_G20_T	Made or received digital payments in the past ...	N

5 rows × 202 columns

In [14]:

```
# See exactly how pandas sees the column names
print(df.columns.tolist())
```

```
['FREQ_x', 'FREQ_LABEL_x', 'REF_AREA_x', 'Country', 'INDICATOR_x', 'INDICATOR_LABEL_
x', 'SEX_x', 'SEX_LABEL_x', 'UNIT_MEASURE_x', 'UNIT_MEASURE_LABEL_x', 'DATABASE_ID_
x', 'DATABASE_ID_LABEL_x', 'UNIT_MULT_x', 'UNIT_MULT_LABEL_x', 'OBS_STATUS_x', 'OBS_
STATUS_LABEL_x', 'OBS_CONF_x', 'OBS_CONF_LABEL_x', '2014_x', '2017_x', '2021_x', '20
22_x', 'STRUCTURE_x', 'STRUCTURE_ID_x', 'ACTION_x', 'FREQ_y', 'REF_AREA_y', 'INDICAT
OR_y', 'SEX_y', 'AGE_x', 'URBANISATION_x', 'UNIT_MEASURE_y', 'COMP_BREAKDOWN_1_x',
'COMP_BREAKDOWN_2_x', 'COMP_BREAKDOWN_3_x', 'AGG_METHOD_x', 'UNIT_TYPE_x', 'DECIMALS
_x', 'DATABASE_ID_y', 'TIME_FORMAT_x', 'COMMENT_TS_x', 'UNIT_MULT_y', 'DATA_SOURCE_
x', 'OBS_CONF_y', 'OBS_STATUS_y', 'FREQ_LABEL_y', 'INDICATOR_LABEL_y', 'SEX_LABEL_
y', 'AGE_LABEL_x', 'URBANISATION_LABEL_x', 'UNIT_MEASURE_LABEL_y', 'COMP_BREAKDOWN_1
_LABEL_x', 'COMP_BREAKDOWN_2_LABEL_x', 'COMP_BREAKDOWN_3_LABEL_x', 'AGG_METHOD_LABEL
_x', 'UNIT_TYPE_LABEL_x', 'DECIMALS_LABEL_x', 'DATABASE_ID_LABEL_y', 'TIME_FORMAT_LA
BEL_x', 'UNIT_MULT_LABEL_y', 'OBS_STATUS_LABEL_y', 'OBS_CONF_LABEL_y', '1990_x', '19
91_x', '1992_x', '1993_x', '1994_x', '1995_x', '1996_x', '1997_x', '1998_x', '1999_
x', '2000_x', '2001_x', '2002_x', '2003_x', '2004_x', '2005_x', '2006_x', '2007_x',
'2008_x', '2009_x', '2010_x', '2011_x', '2012_x', '2013_x', '2014_y', '2015_x', '201
6_x', '2017_y', '2018_x', '2019_x', '2020_x', '2021_y', '2022_y', '2023_x', '2024_
x', 'STRUCTURE_y', 'STRUCTURE_ID_y', 'ACTION_y', 'FREQ', 'REF_AREA', 'INDICATOR', 'S
EX', 'AGE_y', 'URBANISATION_y', 'UNIT_MEASURE', 'COMP_BREAKDOWN_1_y', 'COMP_BREAKDOW
N_2_y', 'COMP_BREAKDOWN_3_y', 'AGG_METHOD_y', 'UNIT_TYPE_y', 'DECIMALS_y', 'DATABASE
_ID', 'TIME_FORMAT_y', 'COMMENT_TS_y', 'UNIT_MULT', 'DATA_SOURCE_y', 'OBS_CONF', 'OB
S_STATUS', 'FREQ_LABEL', 'INDICATOR_LABEL', 'SEX_LABEL', 'AGE_LABEL_y', 'URBANISATIO
N_LABEL_y', 'UNIT_MEASURE_LABEL', 'COMP_BREAKDOWN_1_LABEL_y', 'COMP_BREAKDOWN_2_LABE
L_y', 'COMP_BREAKDOWN_3_LABEL_y', 'AGG_METHOD_LABEL_y', 'UNIT_TYPE_LABEL_y', 'DECIMA
LS_LABEL_y', 'DATABASE_ID_LABEL', 'TIME_FORMAT_LABEL_y', 'UNIT_MULT_LABEL', 'OBS_STA
TUS_LABEL', 'OBS_CONF_LABEL', '1960', '1961', '1962', '1963', '1964', '1965', '196
6', '1967', '1968', '1969', '1970', '1971', '1972', '1973', '1974', '1975', '1976',
'1977', '1978', '1979', '1980', '1981', '1982', '1983', '1984', '1985', '1986', '198
7', '1988', '1989', '1990_y', '1991_y', '1992_y', '1993_y', '1994_y', '1995_y', '199
6_y', '1997_y', '1998_y', '1999_y', '2000_y', '2001_y', '2002_y', '2003_y', '2004_
y', '2005_y', '2006_y', '2007_y', '2008_y', '2009_y', '2010_y', '2011_y', '2012_y',
'2013_y', '2014', '2015_y', '2016_y', '2017', '2018_y', '2019_y', '2020_y', '2021',
'2022', '2023_y', '2024_y']
```

In [15]:

```
## Strip spaces first
## To get the latest year column

digital.columns = digital.columns.str.strip()
internet.columns = internet.columns.str.strip()
gdp.columns = gdp.columns.str.strip()

country_col = 'REF_AREA_LABEL'

def latest_year_col(df):
    year_cols = [col for col in df.columns if col.isdigit() or (col.endswith('_y')
    def col_year(c):
        return int(c.split('_')[0]) if '_' in c else int(c)
    return sorted(year_cols, key=col_year)[-1]
```

In [16]:

```
## Let's get the latest columns

digital_col = latest_year_col(digital)
internet_col = latest_year_col(internet)
gdp_col = latest_year_col(gdp)
```

```
print("Using columns:", digital_col, internet_col, gdp_col)
```

```
Using columns: 2022 2024 2024
```

In [17]:
```
## Let's extract needed data

digital_latest = digital[[country_col, digital_col]].rename(columns={
    country_col: 'Country',
    digital_col: 'Digital Payments'
})

internet_latest = internet[[country_col, internet_col]].rename(columns={
    country_col: 'Country',
    internet_col: 'Internet Users'
})

gdp_latest = gdp[[country_col, gdp_col]].rename(columns={
    country_col: 'Country',
    gdp_col: 'GDP per Capita'
})
```

In [18]:
```
## Let's remove duplicates and merge

digital_latest = digital_latest.drop_duplicates(subset='Country')
internet_latest = internet_latest.drop_duplicates(subset='Country')
gdp_latest = gdp_latest.drop_duplicates(subset='Country')

df_final = digital_latest.merge(internet_latest, on='Country')
df_final = df_final.merge(gdp_latest, on='Country')

print(df_final.head())
```

```
                Country  Digital Payments  Internet Users  GDP per Capita
0           Afghanistan               NaN             NaN             NaN
1                Angola               NaN             NaN    1.009989e+11
2               Albania               NaN             NaN    2.704643e+10
3  United Arab Emirates               NaN        100.0000    5.523248e+11
4             Argentina               NaN         89.6672    6.383655e+11
```

In [19]:
```
## Let's show omly African countries

africa_countries = [
    "Nigeria", "Kenya", "Ghana", "South Africa", "Egypt",
    "Morocco", "Uganda", "Tanzania", "Rwanda"
]
digital = digital[digital["REF_AREA_LABEL"].isin(africa_countries)]
internet = internet[internet["REF_AREA_LABEL"].isin(africa_countries)]
gdp = gdp[gdp["REF_AREA_LABEL"].isin(africa_countries)]
```

In [20]:
```
digital = digital.groupby("REF_AREA_LABEL").last().reset_index()
internet = internet.groupby("REF_AREA_LABEL").last().reset_index()
gdp = gdp.groupby("REF_AREA_LABEL").last().reset_index()
```

In [21]:
```
year = '2022'
```

In [22]:
```
digital_latest = digital[['REF_AREA_LABEL', year]].rename(columns={year:'Digital Pa
internet_latest = internet[['REF_AREA_LABEL', year]].rename(columns={year:'Internet
gdp_latest = gdp[['REF_AREA_LABEL', year]].rename(columns={year:'GDP per Capita'})
```

In [23]:
```
df_final = digital_latest.merge(internet_latest, on='REF_AREA_LABEL', how='inner')
df_final = df_final.merge(gdp_latest, on='REF_AREA_LABEL', how='inner')
```

In [24]:
```
## Let's drop any remaining NaN

df_final = df_final.dropna()
```

In [25]:
```
print(df_final)
```

```
Empty DataFrame
Columns: [REF_AREA_LABEL, Digital Payments, Internet Users, GDP per Capita]
Index: []
```

In [26]:
```
print(digital["REF_AREA_LABEL"].unique())
print(internet["REF_AREA_LABEL"].unique())
print(gdp["REF_AREA_LABEL"].unique())
```

```
['Ghana' 'Kenya' 'Morocco' 'Nigeria' 'Rwanda' 'South Africa' 'Tanzania'
 'Uganda']
['Ghana' 'Kenya' 'Morocco' 'Nigeria' 'Rwanda' 'South Africa' 'Tanzania'
 'Uganda']
['Ghana' 'Kenya' 'Morocco' 'Nigeria' 'Rwanda' 'South Africa' 'Tanzania'
 'Uganda']
```

In [27]:
```
print(digital.columns)
print(internet.columns)
print(gdp.columns)
```

```
Index(['REF_AREA_LABEL', 'FREQ', 'FREQ_LABEL', 'REF_AREA', 'INDICATOR',
       'INDICATOR_LABEL', 'SEX', 'SEX_LABEL', 'UNIT_MEASURE',
       'UNIT_MEASURE_LABEL', 'DATABASE_ID', 'DATABASE_ID_LABEL', 'UNIT_MULT',
       'UNIT_MULT_LABEL', 'OBS_STATUS', 'OBS_STATUS_LABEL', 'OBS_CONF',
       'OBS_CONF_LABEL', '2014', '2017', '2021', '2022'],
      dtype='object')
Index(['REF_AREA_LABEL', 'STRUCTURE', 'STRUCTURE_ID', 'ACTION', 'FREQ',
       'REF_AREA', 'INDICATOR', 'SEX', 'AGE', 'URBANISATION', 'UNIT_MEASURE',
       'COMP_BREAKDOWN_1', 'COMP_BREAKDOWN_2', 'COMP_BREAKDOWN_3',
       'AGG_METHOD', 'UNIT_TYPE', 'DECIMALS', 'DATABASE_ID', 'TIME_FORMAT',
       'COMMENT_TS', 'UNIT_MULT', 'DATA_SOURCE', 'OBS_CONF', 'OBS_STATUS',
       'FREQ_LABEL', 'INDICATOR_LABEL', 'SEX_LABEL', 'AGE_LABEL',
       'URBANISATION_LABEL', 'UNIT_MEASURE_LABEL', 'COMP_BREAKDOWN_1_LABEL',
       'COMP_BREAKDOWN_2_LABEL', 'COMP_BREAKDOWN_3_LABEL', 'AGG_METHOD_LABEL',
       'UNIT_TYPE_LABEL', 'DECIMALS_LABEL', 'DATABASE_ID_LABEL',
       'TIME_FORMAT_LABEL', 'UNIT_MULT_LABEL', 'OBS_STATUS_LABEL',
       'OBS_CONF_LABEL', '1990', '1991', '1992', '1993', '1994', '1995',
       '1996', '1997', '1998', '1999', '2000', '2001', '2002', '2003', '2004',
       '2005', '2006', '2007', '2008', '2009', '2010', '2011', '2012', '2013',
       '2014', '2015', '2016', '2017', '2018', '2019', '2020', '2021', '2022',
       '2023', '2024'],
      dtype='object')
Index(['REF_AREA_LABEL', 'STRUCTURE', 'STRUCTURE_ID', 'ACTION', 'FREQ',
       'REF_AREA', 'INDICATOR', 'SEX', 'AGE', 'URBANISATION',
       ...
       '2015', '2016', '2017', '2018', '2019', '2020', '2021', '2022', '2023',
       '2024'],
      dtype='object', length=106)
```

In [28]:

```python
## Let's Select correct columns

digital_latest = digital[['REF_AREA_LABEL', '2022']].rename(columns={
    'REF_AREA_LABEL': 'Country',
    '2022': 'Digital Payments'
})

internet_latest = internet[['REF_AREA_LABEL', '2022']].rename(columns={
    'REF_AREA_LABEL': 'Country',
    '2022': 'Internet Users'
})

gdp_latest = gdp[['REF_AREA_LABEL', '2022']].rename(columns={
    'REF_AREA_LABEL': 'Country',
    '2022': 'GDP per Capita'
})
```

In [29]:

```python
## Let's Remove duplicates & Merge using INNER JOIN

digital_latest = digital_latest.drop_duplicates(subset='Country')
internet_latest = internet_latest.drop_duplicates(subset='Country')
gdp_latest = gdp_latest.drop_duplicates(subset='Country')

df_final = digital_latest.merge(internet_latest, on='Country', how='inner')
df_final = df_final.merge(gdp_latest, on='Country', how='inner')
```

In [30]:
```
## Let's Drop missing values

df_final = df_final.dropna()

print(df_final)
```

```
Empty DataFrame
Columns: [Country, Digital Payments, Internet Users, GDP per Capita]
Index: []
```

In [31]:
```
print(digital[['REF_AREA_LABEL', '2022']])
print(internet[['REF_AREA_LABEL', '2022']])
print(gdp[['REF_AREA_LABEL', '2022']])
```

```
  REF_AREA_LABEL  2022
0          Ghana   NaN
1          Kenya   NaN
2        Morocco   NaN
3        Nigeria   NaN
4         Rwanda   NaN
5   South Africa   NaN
6       Tanzania   NaN
7         Uganda   NaN
  REF_AREA_LABEL     2022
0          Ghana  69.3607
1          Kenya      NaN
2        Morocco  89.9000
3        Nigeria  37.6975
4         Rwanda  31.2252
5   South Africa  75.4556
6       Tanzania  22.1010
7         Uganda  11.1246
  REF_AREA_LABEL          2022
0          Ghana  7.391900e+10
1          Kenya  1.144490e+11
2        Morocco  1.312453e+11
3        Nigeria  6.469503e+11
4         Rwanda  1.331616e+10
5   South Africa  4.075960e+11
6       Tanzania  7.574912e+10
7         Uganda  4.556533e+10
```

In [32]:
```
## Let's check the year that has data

digital[['2014','2017','2021','2022']]
```

Out[32]:

	2014	2017	2021	2022
0	25.60	49.47	65.57	NaN
1	69.44	78.96	77.56	NaN
2	NaN	16.71	30.30	NaN
3	37.02	29.66	33.74	NaN
4	27.99	38.92	NaN	NaN
5	65.99	60.11	80.81	NaN
6	35.54	42.99	50.22	NaN
7	40.47	54.69	62.58	NaN

In [33]:

```
## Since year 2017 has complete data, we'll make use of that..

year = "2017"

digital_latest = digital[["REF_AREA_LABEL", year]].rename(columns = {
    "REF_AREA_LABEL" : "Country",
    year : "Digital Payments"
})

internet_latest = internet[["REF_AREA_LABEL", year]].rename(columns = {
    "REF_AREA_LABEL" : "Country",
    year : "Internet Users"
})

gdp_latest = gdp[["REF_AREA_LABEL", year]].rename(columns = {
    "REF_AREA_LABEL" : "Country",
    year : "GDP per Capita"
})
```

In [34]:

```
## Let's merge and clean the Dataset

df_final = digital_latest.merge(internet_latest, on = "Country", how = "inner")
df_final = df_final.merge(gdp_latest, on = "Country", how = "inner")

df_final = df_final.dropna()

print(df_final)
```

```
       Country  Digital Payments  Internet Users  GDP per Capita
0        Ghana             49.47         37.8842    6.038541e+10
1        Kenya             78.96         17.8271    8.203651e+10
2      Morocco             16.71         61.7622    1.185406e+11
3      Nigeria             29.66         25.9000    3.757457e+11
4       Rwanda             38.92         17.2000    9.252834e+09
5 South Africa             60.11         56.1674    3.814488e+11
6     Tanzania             42.99         14.4634    5.327488e+10
```

In [35]:

```
!pip install wbgapi pandas
```

```
Requirement already satisfied: wbgapi in c:\users\justdamad\appdata\local\programs\p
ython\python314\lib\site-packages (1.0.14)
Requirement already satisfied: pandas in c:\users\justdamad\appdata\local\programs\p
ython\python314\lib\site-packages (2.3.3)
Requirement already satisfied: requests in c:\users\justdamad\appdata\local\programs
\python\python314\lib\site-packages (from wbgapi) (2.32.5)
Requirement already satisfied: PyYAML in c:\users\justdamad\appdata\local\programs\p
ython\python314\lib\site-packages (from wbgapi) (6.0.3)
Requirement already satisfied: tabulate in c:\users\justdamad\appdata\local\programs
\python\python314\lib\site-packages (from wbgapi) (0.10.0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\justdamad\appdata\local\pro
grams\python\python314\lib\site-packages (from pandas) (2.3.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\justdamad\appdata
\local\programs\python\python314\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\justdamad\appdata\local\prog
rams\python\python314\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\justdamad\appdata\local\pr
ograms\python\python314\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\justdamad\appdata\local\programs
\python\python314\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\justdamad\appdat
a\local\programs\python\python314\lib\site-packages (from requests->wbgapi) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\justdamad\appdata\local\prog
rams\python\python314\lib\site-packages (from requests->wbgapi) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\justdamad\appdata\loca
l\programs\python\python314\lib\site-packages (from requests->wbgapi) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\justdamad\appdata\loca
l\programs\python\python314\lib\site-packages (from requests->wbgapi) (2025.10.5)
```

```
[notice] A new release of pip is available: 25.2 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [36]:
```
import wbgapi as wb
import pandas as pd
```

```
C:\Users\JustDamad\AppData\Local\Programs\Python\Python314\Lib\site-packages\request
s\__init__.py:113: RequestsDependencyWarning: urllib3 (2.5.0) or chardet (7.2.0)/cha
rset_normalizer (3.4.4) doesn't match a supported version!
  warnings.warn(
```

In [37]:
```
countries = ["NGA", "GHA", "KEN", "MAR", "RWA"]
years = range(2010, 2023)
```

In [38]:
```
import requests
import pandas as pd

## Let's fetch and clean data

def fetch_wb_data(indicator):
    url = f"https://api.worldbank.org/v2/country/NGA;GHA;KEN;MAR;RWA/indicator/{ind

    response = requests.get(url)
    data = response.json()

    df = pd.json_normalize(data[1])

    df = df[['country.value', 'date', 'value']]
```

```
    return df

## Let's Fetch datasets

gdp = fetch_wb_data('NY.GDP.MKTP.CD')
internet = fetch_wb_data('IT.NET.USER.ZS')

## Let's Rename columns

gdp.columns = ['Country', 'Year', 'GDP']
internet.columns = ['Country', 'Year', 'Internet_Users']

## Let's Merge

data = pd.merge(gdp, internet, on=['Country', 'Year'])

## Now we can  Clean the Data

data['Year'] = data['Year'].astype(int)
data = data.dropna()

data.head()
```

Out[38]:

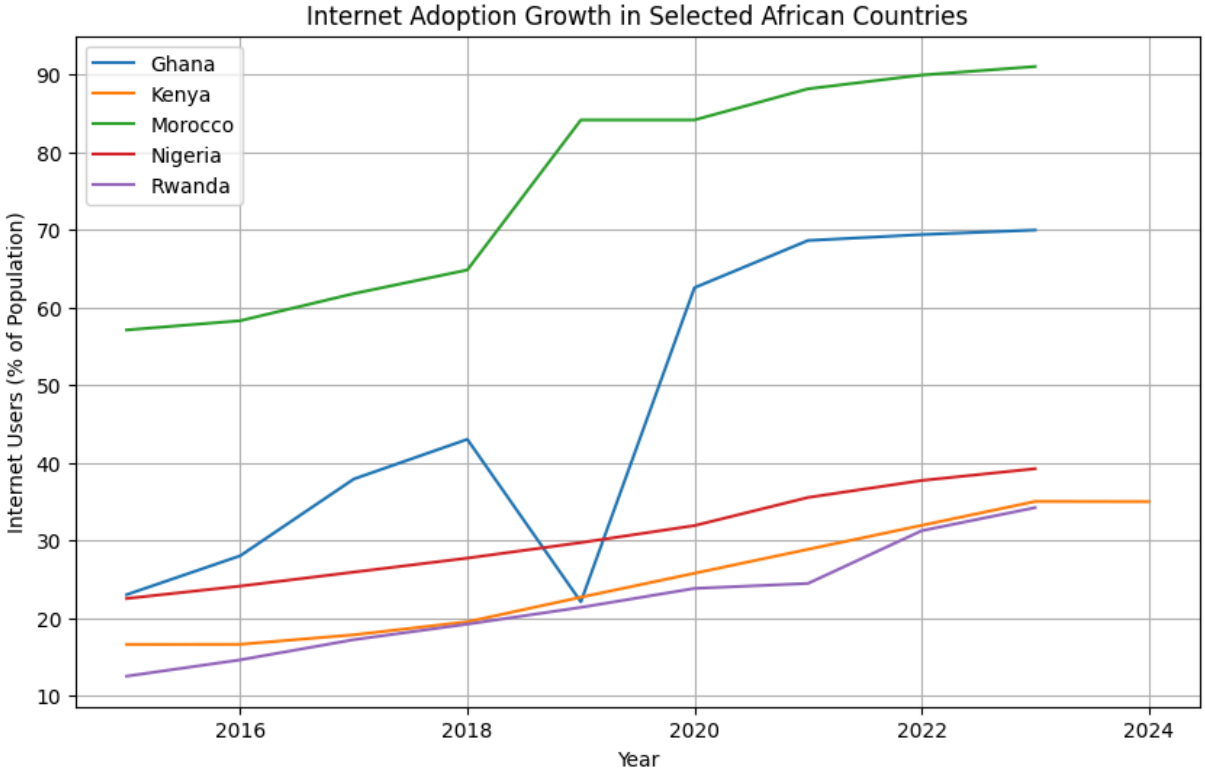
	Country	Year	GDP	Internet_Users
2	Ghana	2023	8.054715e+10	69.9445
3	Ghana	2022	7.391900e+10	69.3607
4	Ghana	2021	7.951420e+10	68.6000
5	Ghana	2020	7.000824e+10	62.5176
6	Ghana	2019	6.835263e+10	22.0732

VISUALIZATION

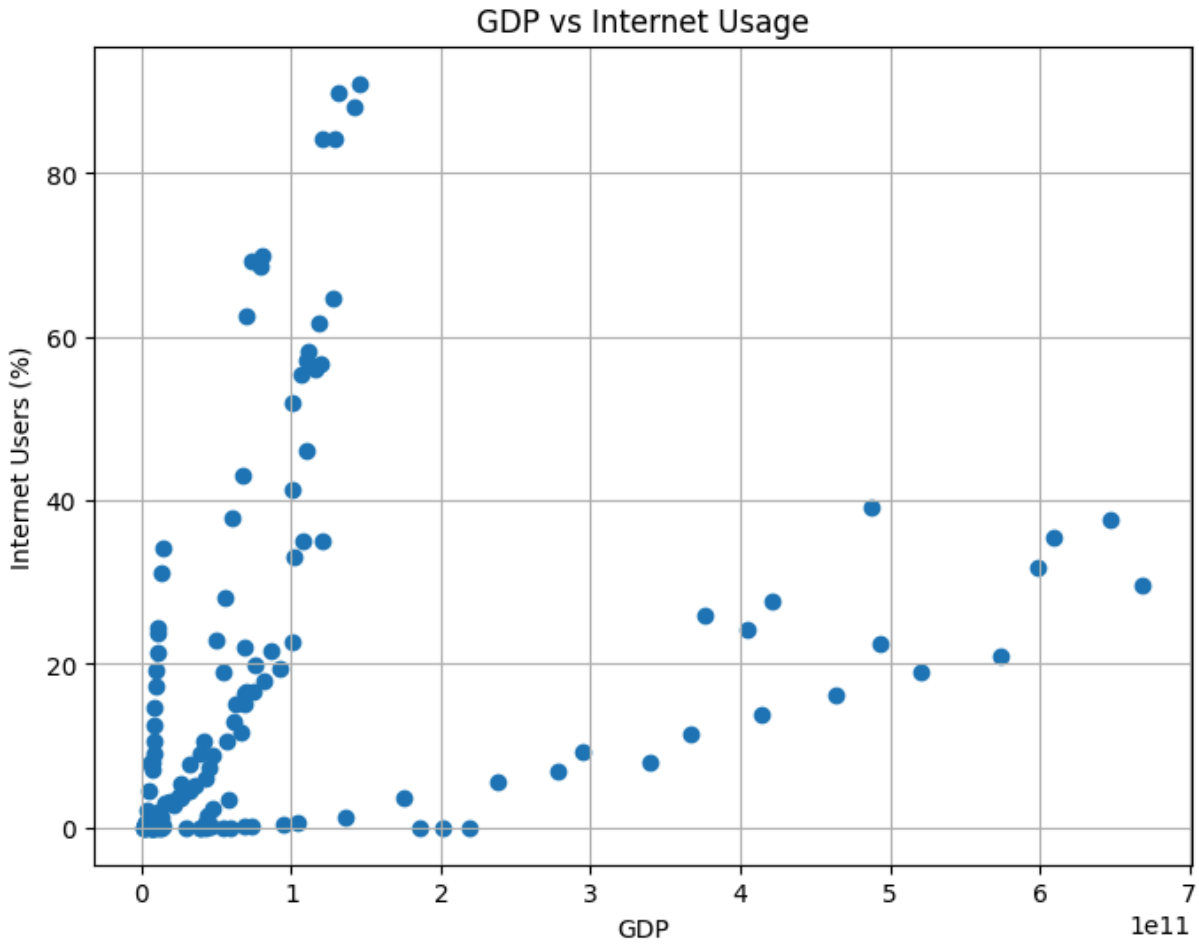
In [39]:

```
import matplotlib.pyplot as plt

## Let's Filter recent years for clarity


df = data[data['Year'] >= 2015]

## Let's  Plot

plt.figure(figsize=(10,6))

for country in df['Country'].unique():
    subset = df[df['Country'] == country]
    plt.plot(subset['Year'], subset['Internet_Users'], label=country)

plt.title("Internet Adoption Growth in Selected African Countries")
plt.xlabel("Year")
plt.ylabel("Internet Users (% of Population)")
```

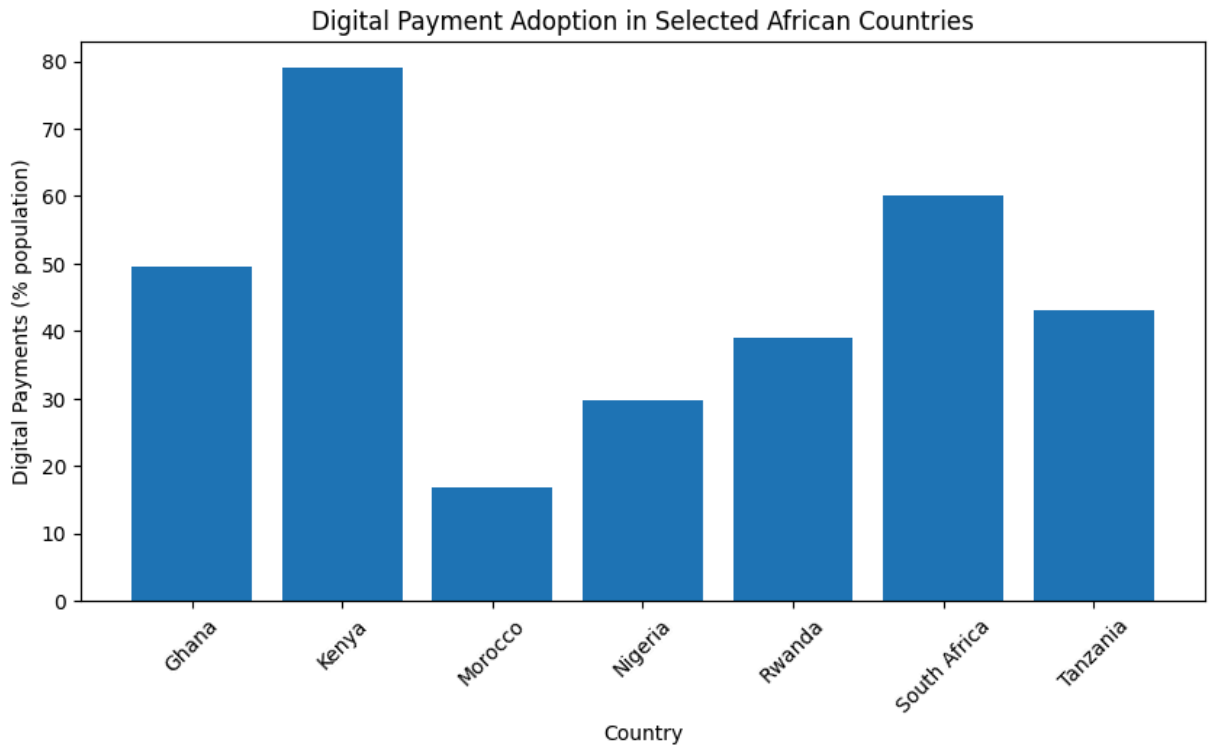
```
plt.legend()
plt.grid()

plt.show()
```



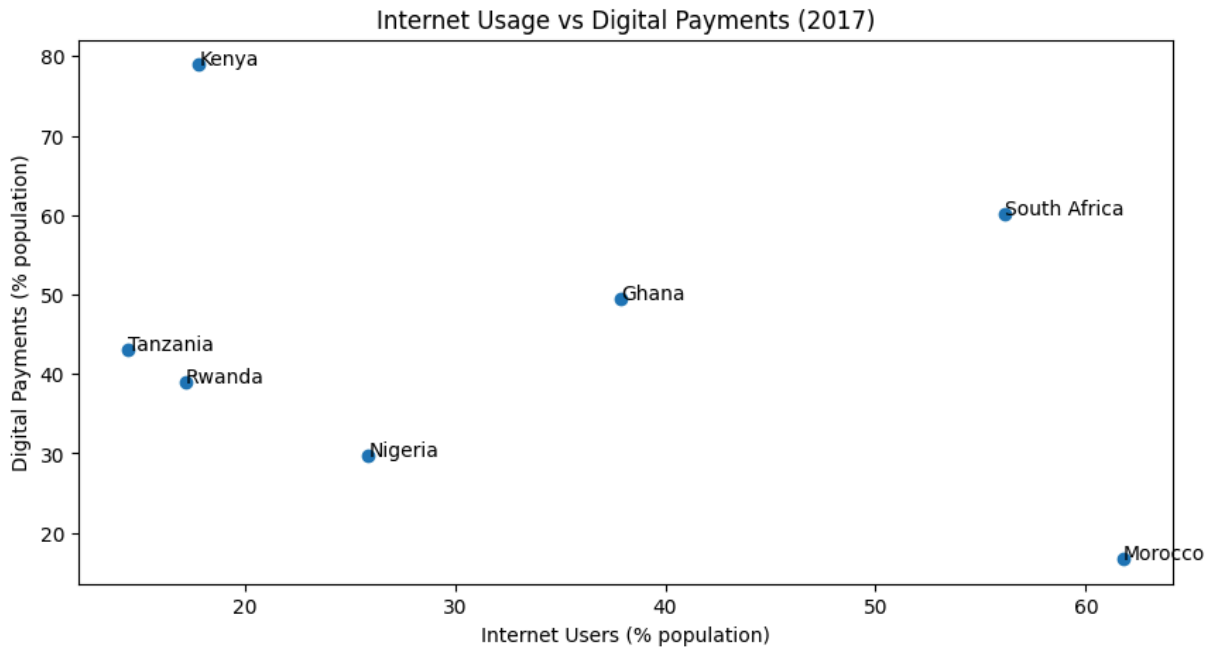
In [40]:
```
## GDP vs Internet scatter

plt.figure(figsize=(8,6))

plt.scatter(data['GDP'], data['Internet_Users'])

plt.title("GDP vs Internet Usage")
plt.xlabel("GDP")
plt.ylabel("Internet Users (%)")

plt.grid()
plt.show()
```



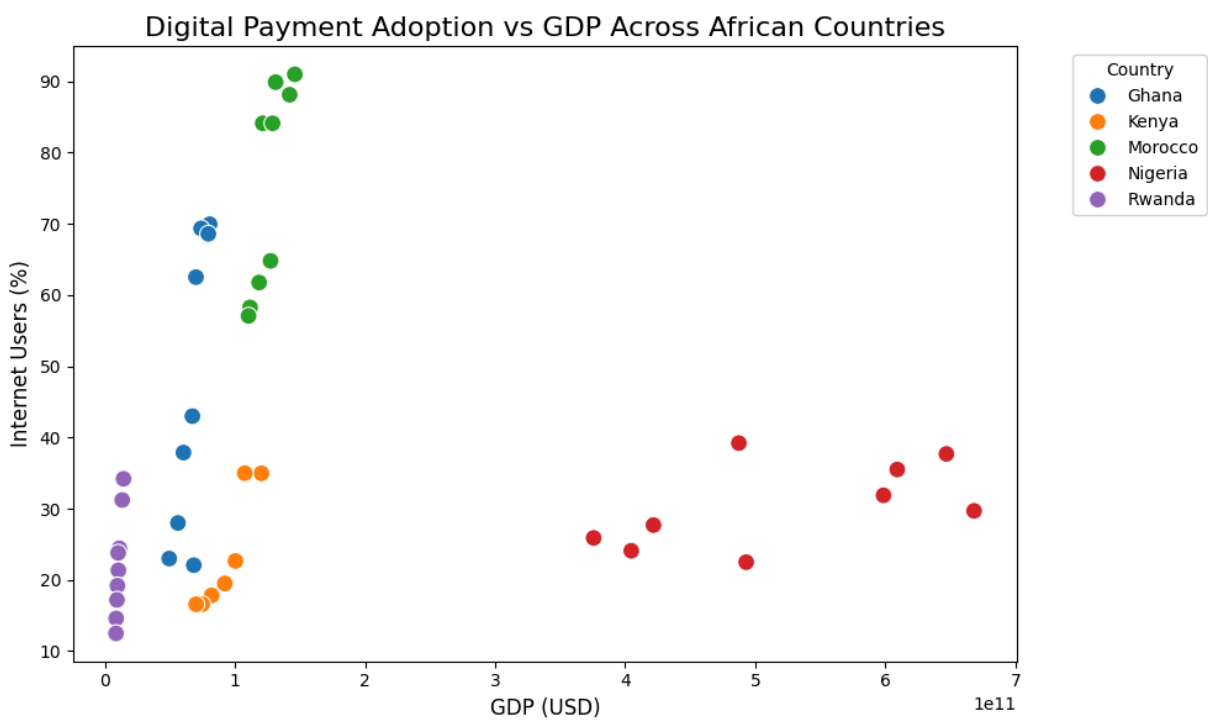
In [41]:
```
## Let's plot a Bar Chart

import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))
plt.bar(df_final['Country'], df_final['Digital Payments'])
plt.title("Digital Payment Adoption in Selected African Countries")
plt.xlabel("Country")
plt.ylabel("Digital Payments (% population)")
plt.xticks(rotation=45)
plt.show()
```



In [42]:
```
## Let's plot a Scatter Plot
## add country labels

plt.figure(figsize = (10,5))
plt.scatter(df_final["Internet Users"] , df_final["Digital Payments"])

for i in range(len(df_final)):
    plt.text(df_final["Internet Users"][i], df_final["Digital Payments"][i], df_fin

plt.title("Internet Usage vs Digital Payments (2017)")
plt.xlabel("Internet Users (% population)")
plt.ylabel("Digital Payments (% population)")
plt.show()
```



In [43]:
```
## Let's Save chart as PNG

plt.figure(figsize=(10,6))
sns.scatterplot(data=df, x='GDP', y='Internet_Users', hue='Country', s=100)
plt.title("Digital Payment Adoption vs GDP Across African Countries", fontsize=16)
plt.xlabel("GDP (USD)", fontsize=12)
plt.ylabel("Internet Users (%)", fontsize=12)
plt.legend(title='Country', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()


plt.savefig("digital_payments_chart.png", dpi=300)  # high-quality PNG
plt.show()
```



KEY FINDINGS :

- Digital payment addoption in African Countries shows a strong relationship with internet access, but a weaker relationship with GDP.

INTERPRETATION :

- Countries with *higher internet penetration* tend to have *higher digital payment usage*
- However, *wealth* (GDP) alone does not guarantee digital adoption
- This suggests that:
  - Digital infrastructure matters more than economic size

CONCLUSION

- My analysis revealed that digital payment adoption in African economies is more strongly influenced by internet accessibility than by GDP levels, highlighting the critical role of digital infrastructure in driving financial inclusion.

SIGNED.

- FinTech Data Analyst.

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